What I Know Now That I Wished I Knew as a Postdoc

Dave Bacon

Who is this Joker?

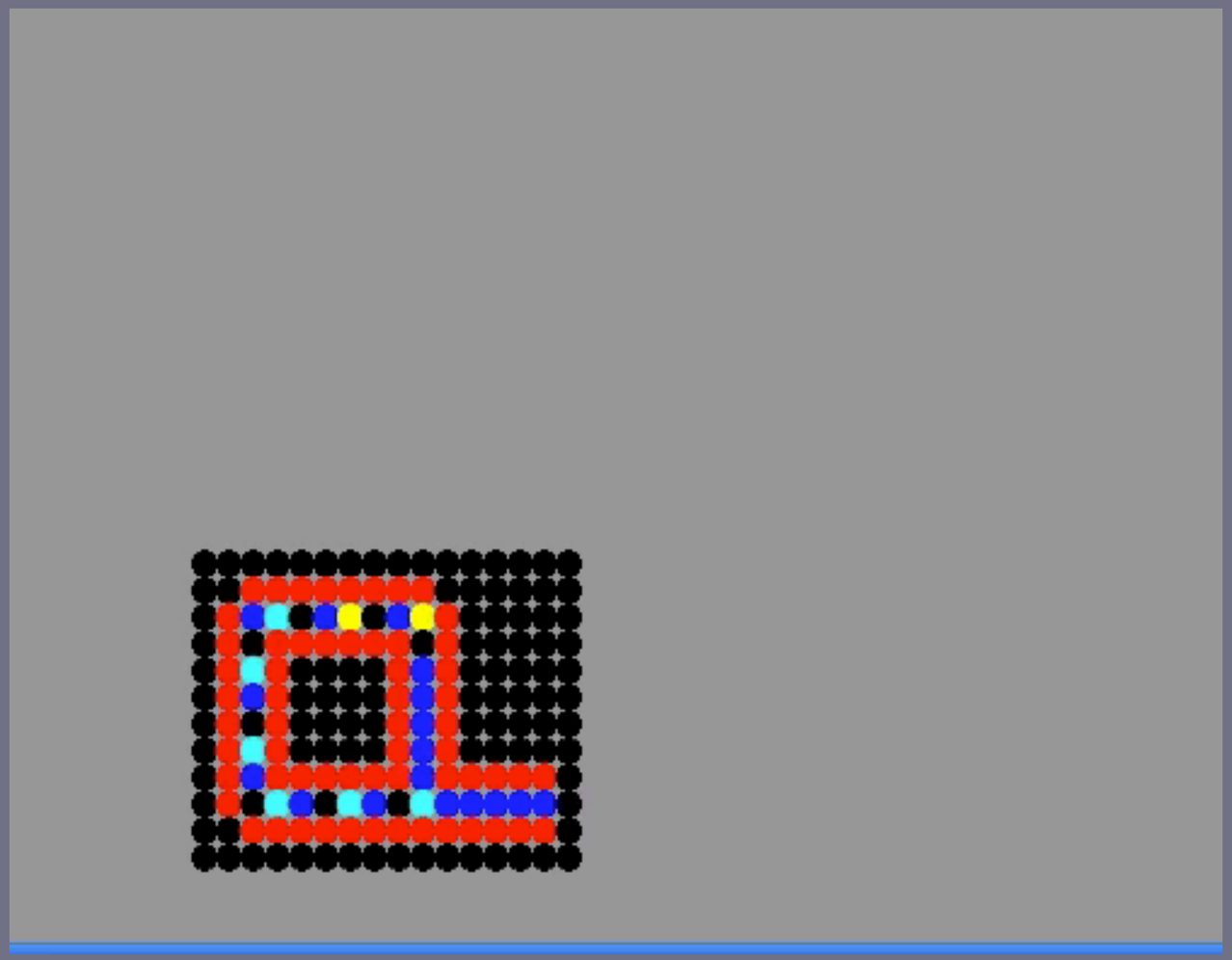
Caltech — Postdoc

SANTA FE INSTITUTE — Postdoc

University of Washington Computer Science & Engineering — Research Prof

Google — Software Engineer?!?!

What Has He Been Doing?



Mantra of a child of the personal computing era

The Universe is a Computer

A Hitch

“Nature isn't classical, dammit, and if you want to make a simulation of nature, you'd better **make it quantum mechanical**, and by golly it's a wonderful problem, because it doesn't look so easy.” - Richard Feynman

Quantum Computers

Classical computers can't* efficiently simulate quantum mechanical systems

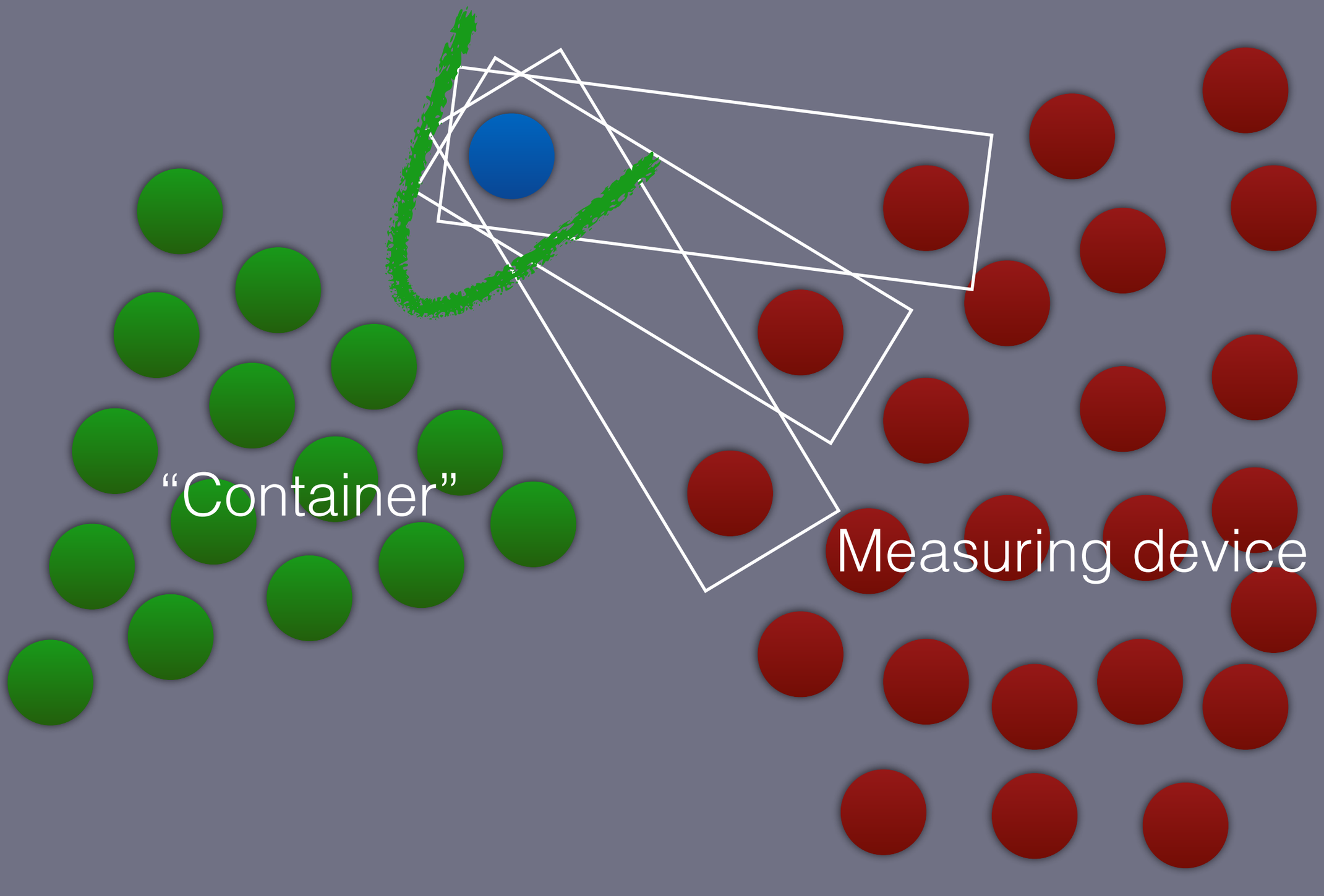
Quantum computers can efficiently solve* problems that classical computers cannot

Where's My Quantum Computer?

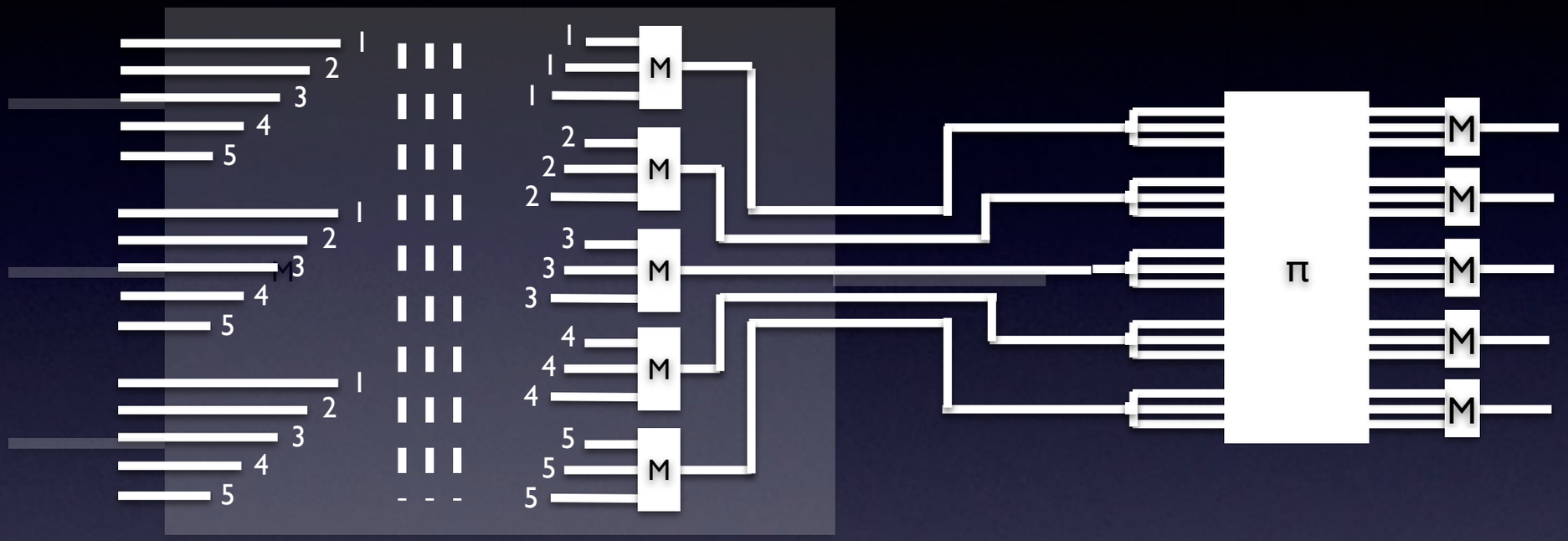
Quantum

The Universe is a Computer

No Qubit is an Island



But We Can Compute



gate error correction

Theorem (von Neumann): A circuit of g gates that fail with probability p can be made to fail with probability ε using O(g log (g/ε)) gates, assuming $p<p_{th}$ (vN estimated $p_{th} \sim 1\%$)

It’s Just a Phase

First transistor
Bell Labs 1947

Computing is a phase of mater

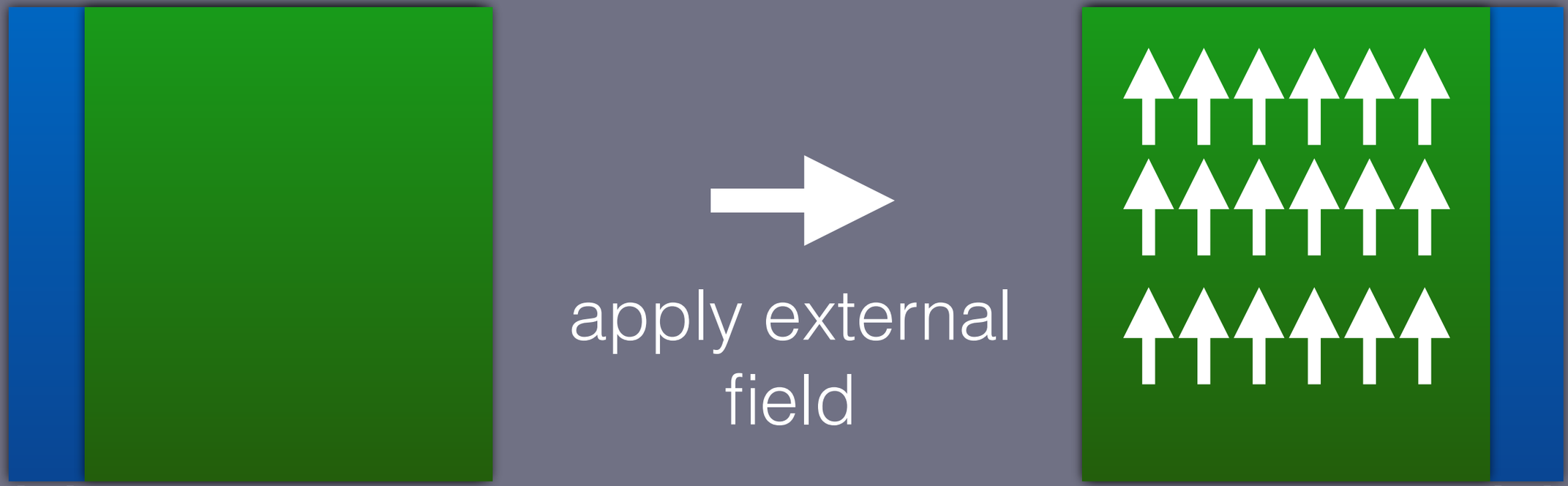
Wherefore Quantum?

Path 1: there is a quantum von Neumann theorem

Path 2: are there quantum computing phases of matter?

Quantum Transistor

[Bacon, Flammia, Crosswhite Phys. Rev. X 3, 021015 (2013)]



Mantra

The anti-Wolfram

Computation is a Phase of Matter

How does computation and memory
arise in a noisy, constantly interacting world?

What other phases of information processing exist?

Postdoc Advice

Caltech — Quantum

SANTA FE INSTITUTE — Quantum

University of Washington Computer Science & Engineering — Quantum

Google — Software Engineer?!?!

Warning!

Advice from the victors is dangerous.

Pursue Artifacts

New

Propagating

Dear Postdoc Dave: “how is what you are working on now going to become an artifact?”

People Help

Dear Past Dave: “Reach out, people want to help”

That Weird Look

New ideas are really hard to understand

….even for smart people

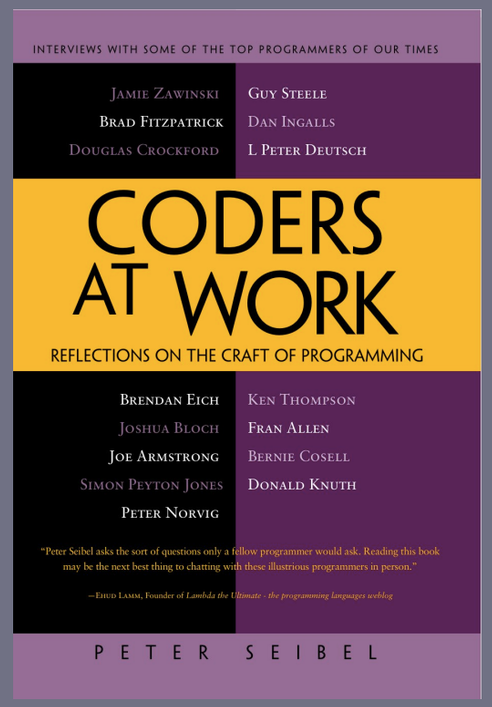
….often the words haven’t been used that way

….and you’ll get looks that seem to be indistinguishable from “you’re crazy”

Past Dave: “Maybe ask them if that look on their face was something they ate?”

Cross Disciplinary Language

Crossing the divide means need to emerge yourself in a new language (and culture).



Sean Hallgren

Past Dave: “Invest (wisely) in learning other disciplines culture and language.”

Paths

Michael Nielsen

Do It

Barbara Terhal

One Final Bit of Advice

Dave: Turn left, not right.

What I Know Now That I Wished I Knew as a Postdoc?

Create, Create, Create!

Information is Fragile

At small scale information is transient

Talking to Past Dave

Quantum Computational Complexity in the Presence of Closed Timelike Curves

Dave Bacon*
Institute for Quantum Information, California Institute of Technology, Pasadena CA 91125, USA and
Department of Physics, California Institute of Technology, Pasadena CA 91125, USA
(Dated: September 29, 2003)